

# Python Tutorial

This PDF is intended for testing a Retrieval-Augmented Generation (RAG) system.

## Chapter 1: Introduction

Python is a high-level, interpreted, general-purpose programming language known for its readability.

## Chapter 2: Variables

Variables store values in memory. Example: `x = 10`, `name = 'Alice'`.

## Chapter 3: Data Types

Python supports `int`, `float`, `str`, `bool`, `list`, `tuple`, `set`, and `dict`.

## Chapter 4: Functions

Functions are reusable blocks of code defined using the `def` keyword.

## Chapter 5: Object-Oriented Programming

Python supports classes, objects, inheritance, encapsulation, polymorphism, and abstraction.

## Chapter 6: Exception Handling

Use `try`, `except`, `else`, and `finally` to handle runtime errors.

## Chapter 7: File Handling

Read files with `open('file.txt', 'r')` and write files with `open('file.txt', 'w')`.

## Chapter 8: Modules

Modules organize reusable code. Example: `import math`.

## Sample Questions

1. What is Python?
2. What are variables?
3. List Python data types.
4. What is a function?
5. Explain OOP.
6. How do you handle exceptions?
7. How do you read a file?
8. What is a module?